50.039 Deep Learning

Anomaly Detection - Depression

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1. Introduction

1.1 Background

The rising global prevalence of mental health disorders has necessitated more robust, objective screening tools, as traditional diagnostic methods often rely on subjective self-reporting that can be prone to recall bias. This project addresses this gap by framing depression detection as a multi-modal anomaly detection task, leveraging the complementary strengths of acoustic features and linguistic patterns. By employing a deep learning architecture that integrates audio signals, capturing prosodic nuances like pitch and latency, with textual data processed through transformer-based models, we can identify subtle deviations from "control" behavioral baselines that might be missed by unimodal systems (Gupta et al., 2023). This multi-modal approach acknowledges that depressive symptoms are rarely confined to a single expressive channel, utilizing fusion techniques to improve classification accuracy and provide a more holistic digital phenotype for early intervention (Zhang et al., 2019; Haque et al., 2018).

1.2 Literature Review

1.2.1 Audio-Based Models

The application of deep learning to acoustic signals has significantly advanced our ability to identify vocal biomarkers of depression. Initial research focused on the use of Convolutional Neural Networks (CNNs) to extract spatial features from visual representations of audio, such as Mel-frequency cepstrograms or spectrograms. Recent studies have demonstrated that 2D-CNNs are particularly effective at identifying local spectral patterns, such as changes in formant frequencies or glottal noise, that correlate with the "flat" or monotonous speech typically observed in depressive subjects (Ishimaru et al., 2022). By treating the audio as an image-like input, these models achieve robust results in differentiating between healthy control groups and those showing signs of clinical disorder.

However, since speech is inherently temporal, recent architectures have increasingly incorporated Long Short-Term Memory (LSTM) networks to capture the long-range dependencies and rhythmic variances in a patient's voice. Unlike standard CNNs, LSTMs are capable of modeling the prosodic "slowness" and increased latency between words that are characteristic of depressive speech (Zhang et al., 2025). Modern hybrid approaches often combine these layers, using CNNs for initial feature

extraction followed by LSTMs to analyze the temporal flow, thereby providing a more nuanced understanding of how vocal characteristics evolve over the duration of a conversation.

1.2.2 Text-Based Models

While audio captures how someone speaks, NLP approaches focus on what is being said, providing a window into the cognitive and emotional state of the individual. Traditional text-based detection relied on sentiment analysis and word-frequency counts, but recent shifts toward deep learning have allowed for the identification of complex linguistic markers, such as increased self-focus through the use of first-person singular pronouns or a higher ratio of negative-to-positive emotional tone (Cariola et al., 2022). These models move beyond simple keyword detection to understand the semantic intent and introspection present in user-generated text, such as diary entries or clinical transcripts (Qasim et al., 2025).

The current state of the art in text-based detection often employs advanced transformer-based architectures like BERT or RoBERTa, which utilize bidirectional context to understand the subtle shifts in meaning within a sentence. Meta-analyses have shown that embedding-based features and deep model architectures significantly outperform shallow machine learning models, as they can represent the subtle linguistic "anomalies" that indicate a deviation from a standard emotional baseline (Qasim et al., 2025). These advancements have proven that text data serves as a natural, highly effective medium for early mental health screening.

1.2.3 Multi-Modal Deep Learning Approach

The convergence of audio and text modalities into a single unified framework addresses the limitations of unimodal systems, which often struggle with the diversity of human expression. By framing depression detection as a multi-modal anomaly detection task, researchers have been able to achieve higher accuracy and robustness; for instance, a system might detect a neutral sentiment in text but identify a depressive vocal "flatness" in the audio, leading to a more accurate classification (Bsoul, 2024). Recent frameworks have utilized techniques of cross-modal attention to fuse these disparate data streams, ensuring that the model learns the complementary relationship between linguistic content and acoustic delivery (Hussain et al., 2026).

1.2.4 Pretrained Feature Extraction Models

To overcome the persistent challenge of limited labeled datasets in the mental health domain, the use of pretrained feature extraction models has become a cornerstone of modern research. Self-supervised models such as WavLM, wav2vec 2.0, and HuBERT are trained on massive amounts of unlabeled speech data, allowing them to learn universal representations that can be fine-tuned for specialized tasks like depression detection (Chen et al., 2022). Similarly, for text, models like RoBERTa provide a deep understanding of language structure that can be adapted to identify depressive markers even with relatively small sample sizes. Leveraging these "foundation" models allows for more sophisticated feature extraction, such as denoising and speaker diarization, resulting in state-of-the-art performance on benchmark datasets like DAIC-WOZ (Gupta et al., 2023).

1.3 Key Takeaways

The synthesis of current research into deep learning for mental health provides a clear roadmap for our project. By viewing depression detection through the lens of anomaly detection, we can move away from rigid, rule-based systems toward a more fluid, nuanced architecture.

The following takeaways represent the foundational pillars of our proposed multi-modal approach:

1. **The Synergy of Audio and Text (Multi-Modality)**
   One of the most critical insights from the literature is that mental health isn't reflected in a single dimension. While text provides a window into a person's cognitive state and semantic intent (e.g., self-focused language), audio captures the physiological and emotional "how" (e.g., monotonous prosody or glottal noise).

   By implementing a **multi-modal architecture**, we aim to mitigate the risk of false negatives. If a patient uses positive words (text) but delivers them with a "flat" or sluggish affect (audio), our model will be equipped to flag this incongruity as a potential anomaly.

2. **Deep Learning for Complex Pattern Recognition**
   Standard machine learning often misses the "why" behind the data. The literature highlights that:
   - **2D-CNNs** are adept at treating spectrograms as images to find local spectral anomalies.
   - **LSTMs** and **Transformers** are essential for capturing the temporal "slowness" and long-range dependencies inherent in depressive speech and thought patterns.

   We will leverage these deep architectures to identify subtle, non-linear markers of depression that traditional sentiment analysis would likely overlook.

3. **Leveraging "Foundation" Models for High Performance**
   Data scarcity is a notorious hurdle in clinical deep learning. The success of self-supervised models like **WavLM** for audio and **RoBERTa** for text suggests that we don't need to start from scratch.

   We will utilize **pretrained feature extraction** as a dual-purpose strategy. This allows us to perform **transfer learning**, where we leverage the massive, pre-existing knowledge these models have of human language and sound.

   Beyond just extracting static features, this approach gives us the flexibility to perform **fine-tuning**. By unfreezing specific layers of these foundation models, we can adapt their internal weights to the specific nuances of depressive speech and text. This ensures our project can achieve state-of-the-art performance even on smaller, specialized datasets, focusing our computational energy on the "downstream" task: the actual classification of depression versus control for anomaly detection.

1.4 Problem Statement

Despite the growing sophistication of automated mental health screenings, current diagnostic frameworks frequently struggle to capture the complex, multifaceted nature of depression. Most existing models treat depression as a static classification problem rather than a dynamic anomaly detection task, failing to account for the significant variance in individual behavioral baselines. Furthermore, unimodal systems, relying solely on text or audio often overlook critical diagnostic cues when symptoms are expressed inconsistently across different communication channels. This results in a "detection gap" where subtle, non-linear indicators of depressive states are dismissed as noise or missed entirely.

There is a critical need for a robust anomaly detection architecture that can integrate acoustic prosody and linguistic semantics to surface clinically relevant deviations from established “control” behaviors. The core challenge lies in developing a deep learning system capable of fusing these disparate data streams to identify when a subject’s expressive patterns diverge from healthy norms. In this context, the proposed approach is intended as a **preliminary assistive tool** for clinicians, providing additional signals to support psychological assessment rather than serving as a standalone diagnostic system. Such augmentation may help mitigate the limitations of subjective self-reporting and the insufficient sensitivity of single-channel automated methods.

2. Data Preparation

2.1 Dataset

To build a robust anomaly detection system, the quality and relevance of the input data are paramount. Our project utilizes the DAIC-WOZ (Distress Analysis Interview Corpus - Wizard of Oz) dataset, a gold-standard benchmark specifically designed for the computational evaluation of psychological distress.

The DAIC-WOZ corpus consists of 189 clinical interviews conducted by an animated virtual interviewer. These sessions are uniquely structured to elicit verbal and non-verbal indicators of anxiety, depression, and PTSD. Each entry in the dataset provides a rich, multi-modal foundation, containing both high-fidelity audio recordings and detailed conversational transcripts.

For simplicity, we adhered to the dataset's pre-defined subsets to ensure comparability with existing literature. The distribution is summarized in the table below:

Split	**Participant Count**	**Depression Count**	**Control Count**	**Ratio out of Total (%)**
Train	107	30	77	56.6
Dev (Validation)	35	12	23	18.5
Test	47	14	33	24.9

Table 1: Dataset Distribution

To enhance robustness, we experimented with mild audio augmentations such as noise injection to account for microphone artifacts and pitch shifting within ±1 semitone. Additionally, spectrogram-based feature extraction was explored as an alternative representation to raw waveform inputs.

2.2 Pre-processing

2.2.1 Audio Pre-processing

To effectively treat depression detection as an anomaly detection task, our audio processing pipeline was designed to isolate the "signal" (the patient's vocal biomarkers) from the "noise" (the interviewer's prompts and environmental silence). This refinement ensures that the deep learning architecture focuses exclusively on the acoustic deviations present in depressive speech.

1. **Diarization and Audio Stitching**
   The raw DAIC-WOZ audio files contain a full dialogue between an automated interviewer and the participant. Because our goal is to identify anomalies within the patient's own behavior, we utilized the provided interview transcripts to perform manual speaker diarization. We extracted only the time-stamped segments where the patient was speaking, discarding all interviewer audio. These isolated patient segments were then stitched together into a single, continuous .wav file per participant. This creates a concentrated "vocal fingerprint" for each individual, removing gaps that might otherwise skew the temporal analysis.

2. **Sliding-Window Segmentation**
   Depressive indicators in speech, such as changes in fundamental frequency or increased pauses, often manifest over several seconds rather than in isolated moments. To capture these rhythmic dependencies while maintaining a robust dataset size, we implemented a sliding-window segmentation strategy:

   - **Window Length (x)**: Each stitched audio file was split into **8-second segments**. This duration was chosen to provide enough context for the WavLM model to extract meaningful paralinguistic features while remaining short enough to capture localized anomalies.

   - **Hop Length (y)**: We applied a **4-second hop**, resulting in a **50% overlap** between consecutive segments. This overlapping approach serves two purposes: it acts as a form of data augmentation and ensures that no critical vocal transitions are lost at the boundaries of a window.

3. **Feature Extraction via Frozen WavLM**
   Rather than relying on traditional spectrograms, we utilized a pretrained **WavLM (base-plus)** model as a high-dimensional feature extractor. WavLM is particularly suited for our task because it was trained with a denoising objective, making it robust against the varied acoustic conditions found in clinical recordings.

For each 8-second segment, the audio was passed through the frozen WavLM layers. We extracted the **768-dimensional latent representations** from the last hidden state and applied mean-pooling over the time dimension. This resulted in a single, dense feature vector per segment, effectively compressing the complex acoustic information into a format that our downstream LSTM and Attention mechanisms can use to identify the subtle "anomalies" characteristic of depression.

Split	**Participant Count**	**Segments**	**Depression Count**	**Control Count**
Train	107	11,362	30	77
Dev (Validation)	35	4,322	12	23
Test	47	5,904	14	33

Table 2: Participant Segments after Feature Extraction

2.2.2 Text Pre-processing

In parallel with our acoustic analysis, the textual component of the DAIC-WOZ dataset provides the semantic context necessary to identify cognitive anomalies. Our text pre-processing pipeline focuses on transforming raw conversational transcripts into structured, context-aware embeddings that highlight linguistic markers of depression.

1. **Transcription and Participant Isolation**
   The initial phase involves transcribing 189 .wav audio files using **OpenAI's Whisper** model. Given that the recordings consist of participant-only audio, the transcription process inherently isolates the patient's speech, providing a focused textual representation of their responses.

   The Whisper "base" model is utilized to balance computational efficiency with transcription accuracy. Each audio file is processed individually, with the resulting text stripped of leading or trailing whitespace before being saved as a separate .txt file in a dedicated transcripts directory. Finally, a sanity check is performed on sample files to ensure the generated text accurately reflects the original audio content.

```
Total transcripts saved: 189

300_P.txt:
  Good. Atlanta, Georgia. My parents are from here. I love it. Like the weather. I like the opportunities. Yes. It's a minute. I'm

301_P.txt:
  I'm doing good. Thank you. I'm from Los Angeles. Oh great. I live in West Los Angeles, the West Side. It's alright. Now I live a

302_P.txt:
  I'm fine, I brought to youself from Los Angeles, California. Apart. Okay. All my families here, friends, mixture of people, and
```

Image 1: Samples of Patient-only Transcripts

2. **Minimal Transcript Cleaning**

Once the transcriptions are complete, a minimal cleaning protocol is applied to the text. The goal is to standardize the data without removing subtle linguistic markers, such as specific word choices or sentence structures; that may serve as useful signals for downstream tasks.

The cleaning process includes the following steps:

- **Normalization**: Unicode is normalized, and Byte Order Mark (BOM) characters are removed to ensure consistent character encoding.
- **Structural Cleanup**: Repeated spaces and line breaks are collapsed to create a continuous text flow.
- **Formatting**: All text is converted to lowercase to maintain consistency across the dataset.
- **Noise Removal**: Non-linguistic junk characters are removed while preserving essential elements like words, numbers, apostrophes, and standard punctuation.
- **Punctuation Spacing**: Spacing around punctuation is normalized to ensure a standard format.

```
Original:

Good. Atlanta, Georgia. My parents are from here. I love it. Like the weather. I like the opportunities. Yes. It's a minute. I'm

Cleaned:

good. atlanta, georgia. my parents are from here. i love it. like the weather. i like the opportunities. yes. it's a minute. i'm
```

Image 2: Sample of Cleaned Transcript - Participant 300

By avoiding heavier processing steps like stemming or stopword removal, the dataset retains its full linguistic richness, providing a robust foundation for subsequent text classification and analysis.

3. **Textual Chunking and Context Windows**
   Because clinical interviews can be lengthy, passing the entire transcript into a transformer model at once would exceed standard token limits and dilute specific emotional markers. To manage this, we implemented a sliding-window chunking strategy:

   - **Chunk Size**: We partitioned the stitched patient text into manageable chunks of **128 tokens**.
   - **Contextual Overlap**: To ensure that the semantic meaning of a sentence isn't lost if it spans across two segments, we utilized a **hop length of 64 tokens**, creating a **50% overlap** between chunks. This ensures that every linguistic transition is captured within the context of the preceding and following statements, allowing the model to detect shifts in sentiment or self-focus more effectively.

4. **Feature Extraction via Mental-RoBERTa**

To move beyond simple keyword counts and capture the nuance of mental health discourse, we employed **Mental-RoBERTa** as our core feature extractor. Unlike a general-purpose language model, Mental-RoBERTa has been specifically pretrained on a massive corpus of mental health-related text, making it uniquely sensitive to the subtle anomalies in vocabulary and syntax associated with psychological distress.

- **Transfer Learning Approach**: We leveraged the model in a frozen state to extract high-dimensional **768-dimensional latent vectors** for each text chunk. By taking the hidden state of the [CLS] token, we obtain a condensed semantic summary of each 128-token window.

- **Adaptability**: This setup allows us to benefit from the model's specialized domain knowledge via transfer learning, while maintaining the flexibility to later fine-tune the classification head on the specific patterns of the DAIC-WOZ dataset.

Split	**Participant Count**	**Chunks**	**Depression Count**	**Control Count**
Train	107	1025	30	77
Dev (Validation)	35	177	12	23
Test	47	260	14	33

Table 3: Participant Chunks after Feature Extraction

3. Model

3.1 Architecture

We initially explored model architectures aimed at predicting PHQ labels through regression, followed by classification for anomaly detection. However, after extensive experimentation, regression-based approaches proved ineffective for this task, leading us to pivot towards more targeted classification loss .

We discussed a few models experimented with a range of models constructed and trained from scratch, to pretrained fine-tune models, to feature extraction from pretrained models to train using transfer learning to a classification head.These models range from audio only features, to using text features.

3.1.1 From scratch CNN Regressor

The model takes as input a log-mel spectrogram with shape (1, 128, 801) where each dimension is channel, mel-frequency bins, and time frames respectively. This representation was chosen as it preserves time-frequency structure of speech while compressing raw waveform into a representation that is easier for a CNN to learn from.

Feature Extractor / Convolution Backbone

The first part of the architecture is a stack of three convolutional blocks. Each block contains:

- Conv2D (Convolution)
- BatchNorm2D (Batch Normalisation)
- ReLU (Rectified Linear Unit)
- MaxPool2D (Maximum Pooling)

The three blocks progress as such:

- Block 1 [1 → 32] channels
- Block 2 [32 → 64] channels
- Block 3 [64 → 128] channels

All convolutions use kernel size of 3, and padding of 1. This keeps spatial structure, while attempting to learn local time-frequency patterns like spectral contours, bursts of energy and prosodic texture.

Compact Feature Representation

After the convolutional blocks, the model applies Adaptive Average Pooling, converting the final feature map into a fixed 4x4 spatial size. This gives a compact and stable representation before regression/classification.

Prediction Head

The pooled representation is then passed through a fully connected regression head.

- Flatten
- Linear(128*4*4 → 256)
- ReLU → Dropout(0.4) → Linear(256 → 64)
- ReLU → Dropout(0.4) → Linear(64 → 1)

The final output is a single scalar, representing the predicted PHQ score.

This was chosen especially as a small 3-block CNN was theoretically simple enough to train on a limited dataset. The fully connected head also allows the extracted acoustic features to be mapped to a continuous PHQ score.

3.1.2 Fine Tuning Wav2Vec Binary Classifier With Augmentation

Input Representation

The model takes as input raw waveform audio from participant-only speech recordings, rather than spectrogram-based features. Each recording is first resampled to 16 kHz, then segmented into fixed-length 8-second windows with a 4-second hop size, producing overlapping segments.

This representation was chosen to allow the model to learn directly from raw speech signals, leveraging pretrained representations instead of relying on hand-crafted spectral features.

Preprocess (Wav cache specific)

Before training, each participant recording is:

- loaded as mono audio
- segmented into overlapping 8-second windows
- stored in a per-participant cache file

This cache stores precomputed segment arrays, allowing repeated training runs to reuse the same segments without recomputing them. This improves both efficiency and reproducibility across experiments.

Data Augmentation

Augmentation is applied only on the training split at the waveform level. The following perturbations are used:

- random gain perturbation
- additive noise
- small speed perturbation
- small time shift

These augmentations were chosen to improve robustness to variations in loudness, background noise, temporal alignment, and speaking rate, while preserving the underlying label semantics.

Feature Extractor

The core of the model is a pretrained transformer-based speech encoder:

- **facebook/wav2vec2-base-960h**

This model converts raw waveform input into high-level contextual speech representations.

Rather than fully retraining the model, partial fine-tuning is applied:

- most lower layers remain frozen
- the feature projection layer is unfrozen
- encoder layer normalization is unfrozen
- the final 2 transformer layers are unfrozen
- the classification head is unfrozen

This setup allows the model to adapt higher-level representations to the task, while retaining general speech features learned during large-scale pretraining.

Classification Head

The model uses a sequence classification head configured with:

- num_labels = 1

The final output is a single logit per segment. This logit is passed through a sigmoid function to obtain a probability representing the confidence of the positive class.

The binary classification target is defined as:

- class 0: PHQ < 10
- class 1: PHQ ≥ 10

The model is trained using a weighted binary cross-entropy loss with logits to account for class imbalance and unequal segment counts across participants.

Although predictions are made at the segment level, ground-truth labels exist at the participant level.

After inference:

- segment-level probabilities are produced
- these are aggregated using weighted averaging
- a final participant-level probability is obtained
- a binary decision is made based on this aggregated score

This ensures that predictions align with the true evaluation objective of classifying participants rather than individual segments.

3.1.3 Transfer Learning

To effectively detect anomalies, our system leverages a mid-fusion ensemble architecture that integrates both linguistic and paralinguistic information. By combining the semantic depth of text with the acoustic nuances of audio, the model can capture subtle indicators of depression that might be missed by a single modality.

The model is structured as a dual-branch network that merges at a central fusion point:

1. **Textual Branch (Mental-RoBERTa)**: We utilize a frozen **Mental-RoBERTa** encoder to process transcript chunks. This model is specifically pre-trained on mental health-related corpora, making it highly sensitive to the semantic markers and emotional vocabulary common in the dataset.

2. **Audio Branch (WavLM)**: Simultaneously, the audio stream is processed via a frozen **WavLM** encoder. This captures essential acoustic features such as pitch, rhythm, and tone, which are critical for identifying the vocal "flatness" or fatigue often associated with depressive states.

3. **Feature Encoding and Attention**: Each modality passes through its own LSTM layer followed by an attention mechanism. This allows the model to weigh specific parts of a sequence, such as a particularly heavy sigh in the audio or a poignant phrase in the text, more heavily than others.

4. **Mid-Fusion Concatenation**: The resulting 64-dimensional text embedding and 128-dimensional audio embedding are concatenated into a single, unified feature vector. This "mid-fusion" approach is superior to early fusion (merging raw data) or late fusion (averaging final scores) because it allows the model to learn the complex correlations between how someone speaks and what they are saying.

5. **MLP Classifier**: The fused vector is passed into a trainable Multi-Layer Perceptron (MLP) consisting of a 128-unit hidden layer with ReLU activation and Dropout for regularization, culminating in a single logit for binary classification.

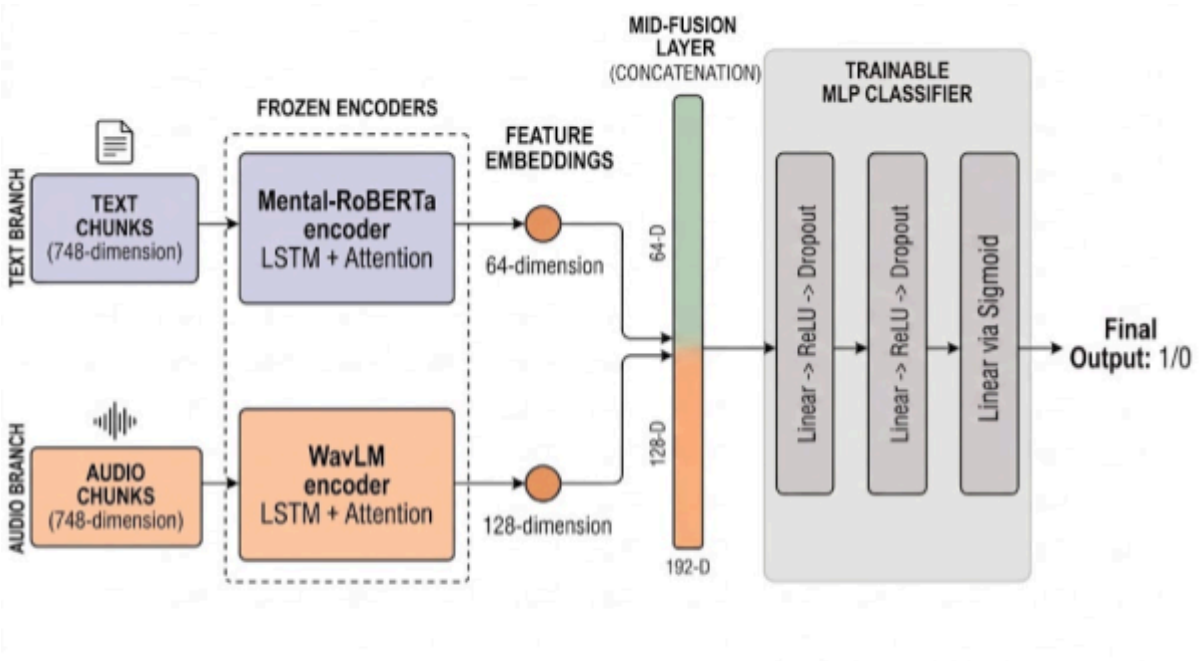

Image 3: Mid-Fusion Ensemble Model

The choice of this architecture is driven by the principle of **Transfer Learning**. Rather than training a model from scratch on a relatively small and imbalanced dataset, we leverage the "deep knowledge" embedded in RoBERTa and WavLM. By freezing these high-level encoders, we reduce the risk of overfitting and ensure that our model benefits from features learned on millions of hours of speech and text data.

Designing the architecture as an ensemble ensures a "fail-safe" mechanism: if a participant's speech is acoustically clear but semantically concerning (or vice versa), the fusion layer can still identify the anomaly. This holistic view is essential for a task as nuanced as mental health monitoring, where the signal is often distributed across different forms of expression.

3.2 Loss

Across the various experiments, the loss function was chosen based on both the data modality and whether the task was approached as regression or classification. Although the problem is inherently a classification task, we initially explored a regression-based formulation, leading to the use of Mean Squared Error, before ultimately transitioning to Binary Cross Entropy for anomaly detection.

3.1.3 Mean Square Error Regression

In the initial formulation of the task, we explored whether the problem could be approached from a more ambitious regression perspective, directly predicting continuous PHQ scores rather than discretizing into classes. This would, in principle, allow the model to capture finer-grained variations in depression severity, beyond a simple binary decision boundary.

To support this objective, the model was trained using Mean Squared Error (MSE) loss, which penalizes the squared difference between predicted and ground-truth scores. Unlike classification losses, which focus on separating classes, MSE encourages the model to learn a continuous mapping from acoustic features to PHQ severity.

Mathematically, the MSE loss is defined as:

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2$$

where:

- N is the number of samples in the batch
- $\hat{y}_i$ is the predicted PHQ score for sample i
- y_i is the ground-truth PHQ score

The squared error formulation places greater emphasis on larger deviations, encouraging the model to avoid significant misestimation of a participant's condition, which is particularly relevant when differences in score correspond to clinically meaningful changes.

However, despite the appeal of this approach, we found that regression-based training did not translate effectively into the anomaly detection objective. In practice, the model struggled to produce outputs that could be reliably thresholded into meaningful control vs depressed distinctions. As a result, we transitioned toward a classification-based framework using Binary Cross Entropy, which allowed the model to learn this decision boundary more directly.

3.2.2 Weighted Binary Cross Entropy Logits Loss

In the context of anomaly detection, the "anomalous" class is, by definition, the minority. During the initial development of our multi-modal ensemble, we utilized a standard **Binary Cross Entropy with Logits Loss (BCE)**. While effective for balanced datasets, this approach treats every misclassification equally. In our case, where the dataset exhibits a significant imbalance between Control and Depressed samples, this led to a model that was naturally biased toward the majority class, often overlooking the critical but rare anomalies, depressed samples, we aimed to detect.

To correct this, we shifted to **Weighted Binary Cross Entropy with Logits Loss**. This modification allows us to assign a higher penalty to errors made on the minority class, effectively "forcing" the model to pay closer attention to anomalous patterns in both the text and audio streams.

Mathematically, Standard BCE calculates loss as:

$$L = -[y \cdot log(\sigma(x)) + (1 - y) \cdot log(1 - \sigma(x))]$$

Where $\sigma(x)$ is the sigmoid function. When we introduce a weight w for the positive class (the anomaly, depressed samples), the loss function becomes:

$$L_{weighted} = -[w \cdot y \cdot log(\sigma(x)) + (1 - y) \cdot log(1 - \sigma(x))]$$

By setting w based on the inverse frequency of the classes, we ensure that the ensemble doesn't simply achieve high accuracy by predicting the majority class every time.

In our ensemble approach, which integrates features from both audio (e.g., prosody, tone) and text (e.g., sentiment, semantics), the weighted loss serves as a critical stabilizer. Because anomalies might be subtle in one modality but glaring in the other, the weighted loss prevents the model from dismissing these "weak but important" signals.

This change has moved our performance metrics away from "inflated accuracy" and toward a more meaningful F1-score and Recall, ensuring that our system is actually reliable for real-world detection where missing a depressed sample is far more costly than a false alarm.

4. Training

4.1 Configuration

The training process for the mid-fusion ensemble model is conducted using a set of predefined configurations to ensure stability and reproducibility before the architecture and optimization parameters undergo fine-tuning. For this anomaly detection task, the model integrates pre-computed text embeddings from a Mental-RoBERTa branch and audio embeddings from a WavLM branch.

The following configurations represent the static foundation of the training pipeline:

- **Training Epochs**: The model was configured to train for a maximum of 500 epochs. To protect against overfitting and unnecessary computation, an early stopping mechanism was integrated with a patience of 50 epochs; this ensured that training would terminate if the validation macro-F1 score failed to show improvement.

- **Optimizer**: We utilized the Adam optimizer for weight updates. Adam was preferred for this anomaly detection task due to its adaptive learning rate capabilities, which are particularly effective when training MLP heads on top of pre-extracted, high-dimensional features from the Mental-RoBERTa and WavLM branches.

- **Loss Function**: The training objective was defined by the Binary Cross Entropy with Logits (BCEWithLogitsLoss). Given that anomaly detection datasets are frequently imbalanced, a positive class weight of approximately 2.57 was applied to the loss function to ensure the model remained sensitive to the minority class.

- **Seed**: To guarantee the reproducibility of the results and ensure that performance gains were due to architectural changes rather than stochastic variance, a global seed of 42 was applied across all random number generators, including NumPy and PyTorch.

4.2 Hyperparameter Tuning

Following the establishment of the fixed baseline, we moved into a systematic hyperparameter tuning phase to refine the performance of the mid-fusion ensemble. Table 4 outlines the predefined search space we explored, which included variations in architectural dimensions, dropout probabilities, and optimization variables like learning rate and weight decay.

To identify the most effective configuration, we narrowed our analysis to the top 10 performing combinations discovered during the grid search. Interestingly, we found ourselves at a technical crossroads: all ten of these top-tier combinations achieved an identical **validation macro-F1 score**. While this indicated a level of stability in the model's performance across certain parameter ranges, it meant that F1 score alone was insufficient for selecting a single "best" model.

Hyperparameter	**Search Space**
hidden_dim	{64, 128, 256}
dropout	0.2 to 0.6
lr	1e-5 to 1e-3
weight_decay	1e-5 to 1e-2
noise_std	0 to 0.1
batch_size	{8, 16, 32}

Table 4: Hyperparameter Search Space

	trial_number	state	val_macro_f1	best_val_loss	hidden_dim	dropout	lr	weight_decay	noise_std	batch_size
0	4	COMPLETE	0.761905	0.926849	128	0.204927	0.000579	0.000120	0.095826	16
1	20	COMPLETE	0.761905	0.931858	64	0.432302	0.000113	0.000224	0.099469	32
2	6	COMPLETE	0.761905	0.932035	128	0.389231	0.000147	0.000019	0.025318	32
3	3	COMPLETE	0.761905	0.933698	64	0.298207	0.000197	0.000339	0.048378	16
4	29	COMPLETE	0.761905	0.933998	64	0.370084	0.000181	0.000810	0.086244	32
5	33	COMPLETE	0.761905	0.938036	128	0.238340	0.000329	0.000063	0.092257	16
6	47	COMPLETE	0.761905	0.941613	256	0.560546	0.000204	0.002010	0.008695	32
7	35	COMPLETE	0.761905	0.941753	128	0.205277	0.000093	0.000167	0.087339	16
8	37	COMPLETE	0.761905	0.941937	64	0.344004	0.000707	0.000360	0.077050	16
9	43	COMPLETE	0.761905	0.943781	64	0.526507	0.000195	0.000524	0.050066	16

Image 4: Top 10 completed trials ranked by Validation Loss after Tie-Break Rule

To resolve this tie-break and select the final production parameters, we introduced a secondary ranking metric. We evaluated the **validation loss** for each of these ten candidates as shown in the Bar Plot. By choosing the combination with the lowest validation loss among those tied for the highest F1 score, we ensured that the final model was not only accurate but also more confident and better calibrated in its predictions.

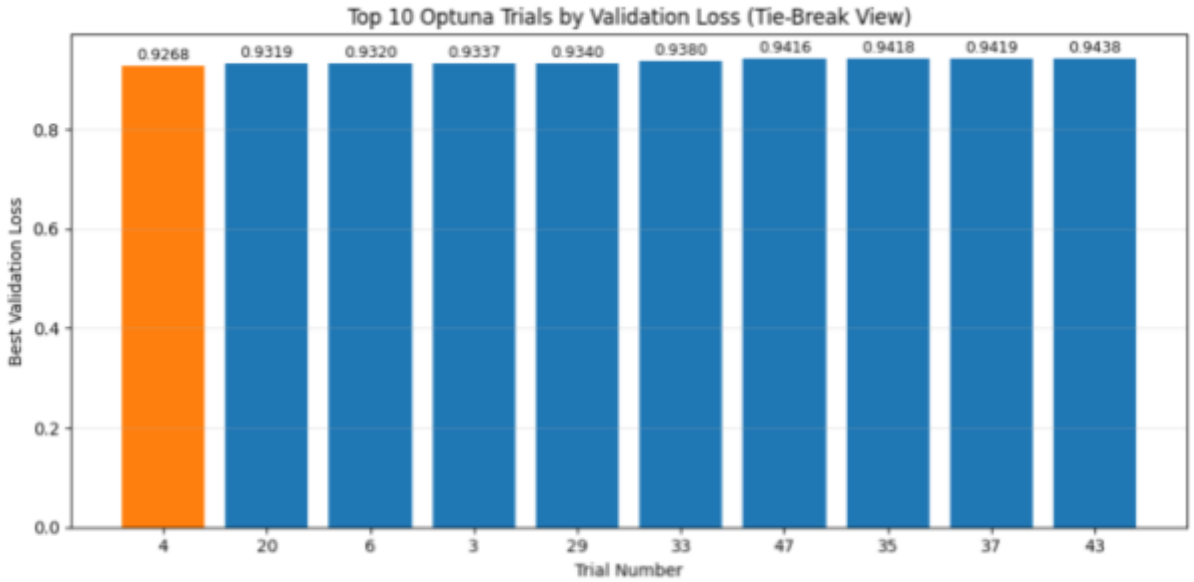

Image 5: Bar Plot of Top 10 trials ranked by Validation Loss

The resulting optimal hyperparameters for the mid-fusion model are:

- **Hidden Dimension**: 128
- **Dropout**: 0.205
- **Learning Rate**: $5.79\ x\ 10^{-4}$
- **Weight Decay**: $1.20\ x\ 10^{-4}$
- **Batch Size**: 16
- **Noise Std Dev**: 0.096

5. Evaluation Results

5.1 Evaluation of Traditional ML to initial DL methods

Model	Rmse	mae	pearson_r
Ridge Regression	67887.0214	9908.2880	0.1606
Elastic Net	69272.7090	10110.4208	0.1606
Support Vector Regression	6.686083	5.290332	0.0451
Random Forest	6.4918	5.3875	0.0540
3 Block CNN	6.3904	5.4863	0.1295

Table 5: Evaluation Results for Machine Learning and 3 block CNN model

As shown in Table 5.1, traditional machine learning models such as Ridge Regression, Elastic Net, and Support Vector Regression exhibited weak performance in terms of Pearson correlation, indicating a limited ability to capture meaningful relationships between the input features and PHQ scores. While these approaches were able to achieve moderate error values, the low correlation suggests that predictions were not well aligned with the underlying structure of the data.

Transitioning to a deep learning approach with the 3-block CNN resulted in a reduction in overall error, achieving the lowest RMSE among the evaluated models. However, the corresponding Pearson correlation remained only modestly improved, and still comparable to traditional methods. This indicates that while the CNN was better at minimizing aggregate error, it did not significantly enhance the model's ability to track the true variation in PHQ scores.

Metric	Score (%)
Test Accuracy	54
Precision (Macro Avg)	48
Recall (Macro Avg)	58
F1-Score (Macro Avg)	50

Table 6: Evaluation Results for Wav2Vec Data Augmented Model

As shown in Table 6, the Wav2Vec-based model achieved only modest performance across all metrics, indicating that relying on a single modality, even with a pretrained architecture, was insufficient for this task. This motivated the exploration of more integrated approaches beyond standalone models.

5.2 Evaluation of Mid-Fusion Ensemble Model

The training evaluation for the mid-fusion ensemble model demonstrates strong performance for the specific task of anomaly detection (identifying indicators of depression). This was achieved by combining the contextual nuance of Mental-RoBERTa text embeddings with the acoustic patterns of WavLM audio embeddings.

The model was tested on a dataset consisting of **47 participants (33 Control, 14 Depressed)**. Using the optimal hyperparameters determined during the tuning phase, the results are as follows:

To truly understand the model's effectiveness, we must look at the standard classification metrics in direct relation to the Confusion Matrix (Figure 6):

Metric	**Score (%)**
Test Accuracy	81
Precision (Macro Avg)	78
Recall (Macro Avg)	82
F1-Score (Macro Avg)	79

Table 7: Evaluation Results for Mid-Fusion Ensemble Model

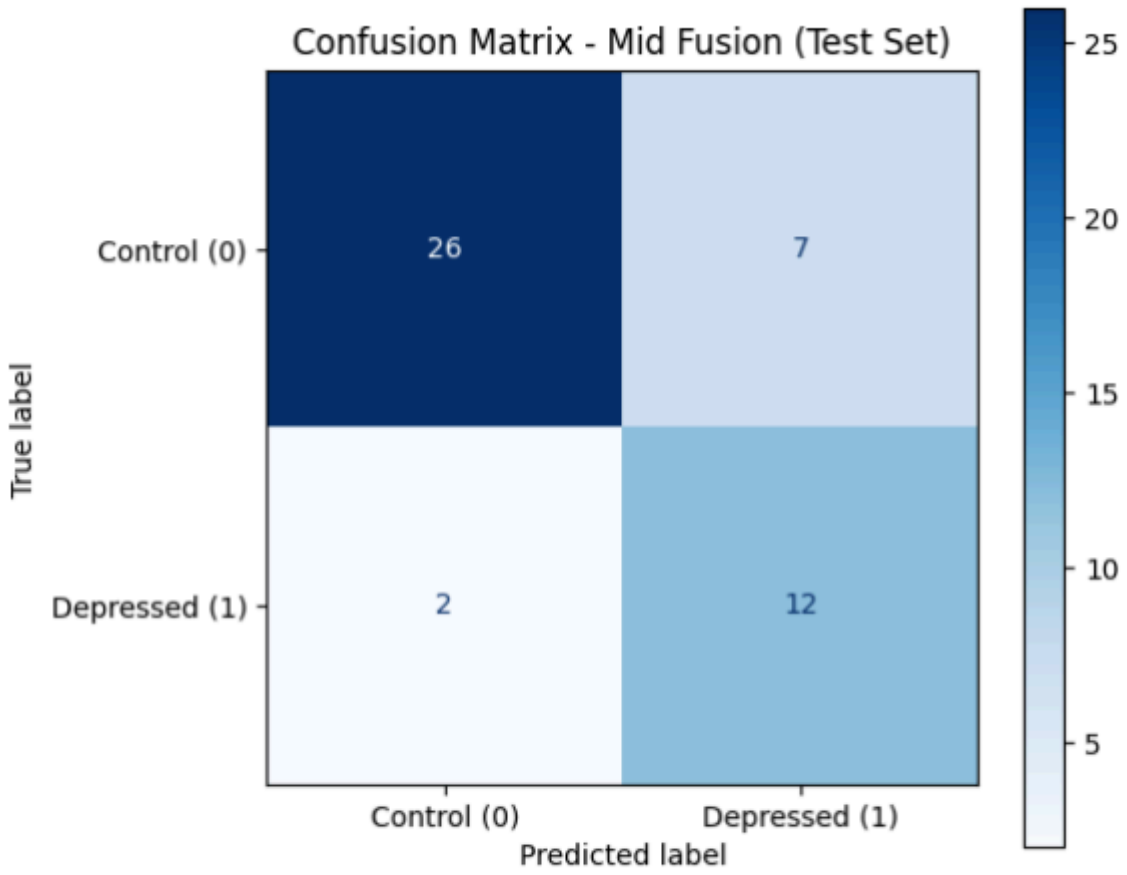

Image 6: Confusion Matrix after Evaluation

The performance metrics and confusion matrix indicate that this ensemble model is a highly capable and reliable tool for anomaly detection, specifically in the context of behavioral health screening. Its reliability is rooted in its prioritization of sensitivity, ensuring that "anomalous" (depressed) signals are rarely missed.

- **Reliability in Anomaly Detection**: In anomaly detection, the "cost" of a false negative is often much higher than a false positive. This model exhibits a Recall of 86% for the depressed class, meaning it successfully captured 12 out of 14 anomalous cases. From a reliability standpoint, this suggests the model has successfully learned the distinctive acoustic and textual patterns, the "anomalies", that separate the depressed class from the control group. With only two missed cases (false negatives), the model demonstrates a high degree of dependability for early-intervention scenarios where identifying every potential risk is the primary objective.

- **Capability as an Ensemble Classifier**: The model's capability is highlighted by its Macro F1-Score of 79%, which confirms it isn't simply guessing based on the majority class. Because the dataset is imbalanced (33 Control vs. 14 Depressed), a naive model might simply predict "Control" every time to achieve high accuracy. Instead, this ensemble demonstrates its sophistication by maintaining an 81% accuracy while still accurately identifying the minority class. While the 63% precision indicates a tendency to over-flag (7 false positives), this is a

common characteristic of robust anomaly detection systems; they are designed to be "cautious," flagging subtle deviations for further human review rather than ignoring them.

5.3 Comparison of Models

The journey to developing a robust classifier for anomaly detection was defined by incremental breakthroughs, as we moved from basic single-modality baselines to a sophisticated multimodal ensemble.

Model	**Modality**	**Accuracy**	**Precision**	**Recall**	**F1-Score**
TF-IDF + Logistic Regression	Text	0.5532	0.6	0.55	0.57
RoBERTa	Text	0.7234	0.78	0.72	0.73
Mental-RoBERTa	Text	0.6596	0.8	0.66	0.67
Mental-RoBERTa + LSTM	Text	0.8085	0.84	0.81	0.82
Wav2vec 2.0	Audio	0.5957	0.59	0.6	0.59
HuBERT	Audio	0.6383	0.63	0.64	0.63
WavLM	Audio	0.7021	0.68	0.7	0.69
WavLM + LSTM	Audio	0.6383	0.63	0.64	0.63
Mid-Fusion Ensemble (Mental-RoBERTa + WavLM)	**Text + Audio**	**0.81**	**0.84**	**0.81**	**0.82**
Early-Fusion Ensemble (Mental-RoBERTa + WavLM)	Text + Audio	0.66	0.65	0.66	0.65

Table 8: Comparison of Models on Test Set

Initially, our experiments confirmed that relying on a single data source, either audio or text, was insufficient for creating a strong enough classifier. Standard approaches like TF-IDF with Logistic Regression or standalone Wav2vec 2.0 struggled to capture the complex, multifaceted nature of behavioral anomalies, yielding F1-scores that hovered between 0.57 and 0.59. These results underscored the reality that depression manifests in both what is said and how it is spoken; ignoring one half of that equation limits the model's diagnostic depth.

Our first major breakthrough came when we pivoted to Mental-RoBERTa for text feature extraction. Unlike general-purpose language models, Mental-RoBERTa is pretrained on domain-specific medical and mental health data, making it far better suited for the nuances of our dataset. We saw a subsequent leap in performance by employing an LSTM layer to learn from these feature embeddings before

classification. This allowed the model to account for the temporal flow of conversation rather than treating words as isolated data points. For the audio modality, a similar iterative process led us to WavLM, which consistently outperformed HuBERT and Wav2vec 2.0, providing the cleanest representation of acoustic anomalies.

With the best-performing feature extractors identified, **Mental-RoBERTa** for text and **WavLM** for audio, we moved into the ensemble phase. We specifically compared **Early-Fusion** (concatenating raw inputs early on) against **Mid-Fusion** (processing features independently before merging them at a deeper layer).

The results were definitive: the **Mid-Fusion Ensemble** emerged as our most capable model, achieving an **Accuracy of 81%** and a **Macro F1-Score of 0.82**. We found that Mid-Fusion was the most appropriate architecture because it facilitates cross-feature interaction. By allowing the model to weigh the importance of specific vocal cues against specific textual themes simultaneously, the performance received a significant boost.

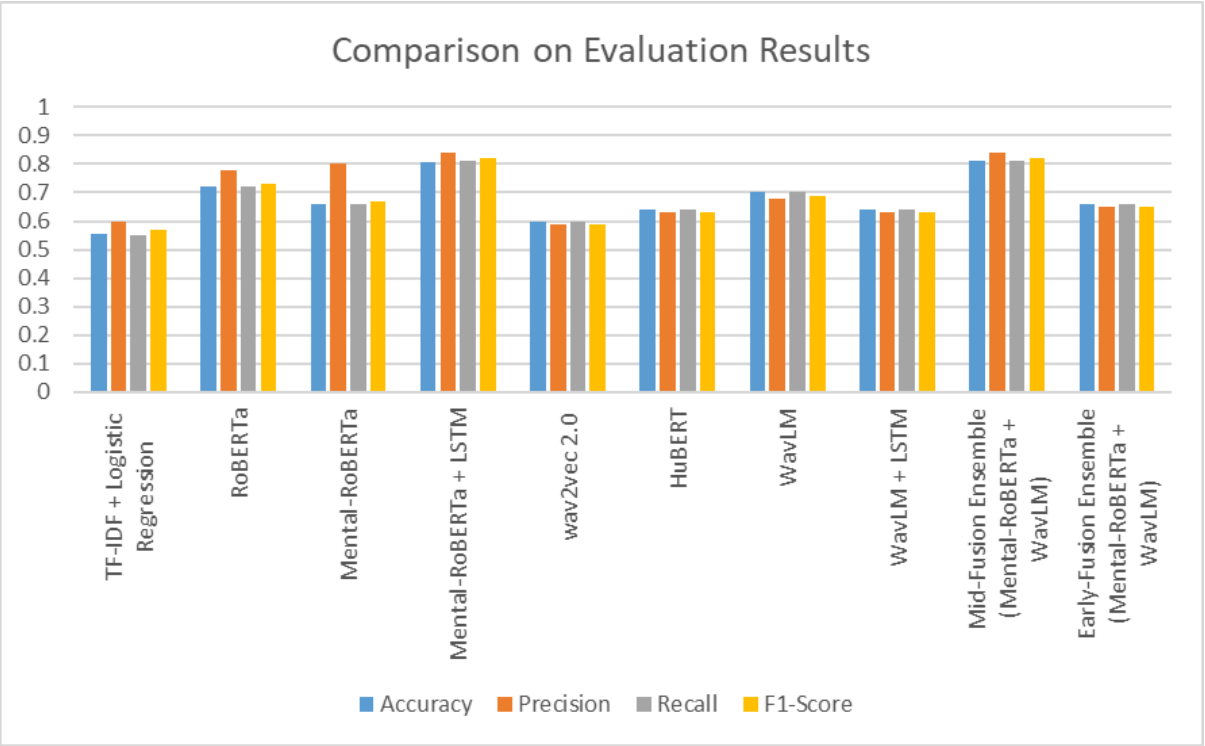

Image 7: Box Plot of Models

To ensure a fair comparison, all models in this study utilized a consistent classification head:

- **Attention Pooling**: To focus on the most relevant parts of the signal.
- **Layer Normalization**: For training stability.
- **MLP Classifier**: Using a sigmoid activation to handle the binary classification between "Control" and "Depressed."

Ultimately, the Mid-Fusion approach proved to be the most reliable for anomaly detection. In a project where depressed samples are the minority, this model's ability to integrate diverse signals

ensures that these critical anomalies are detected with high sensitivity without sacrificing overall precision.

While the performance gain from the ensemble over the best single-modality models was modest, on the order of approximately one percentage point, this improvement is nonetheless meaningful. It suggests that the model is beginning to benefit from cross-modal interactions, even within the constraints of a limited dataset. In particular, the textual modality, powered by Mental-RoBERTa, appears to have reached a relatively strong baseline given the dataset size, whereas the audio modality retains additional headroom. As such, the contribution from WavLM is likely to become more pronounced with increased data availability, where richer acoustic patterns can be more effectively learned. This indicates that the observed gains from Mid-Fusion are not merely incremental, but reflective of a scaling trend that would favor multi-modal approaches as dataset size grows.

6. Reproducing

To ensure that our findings are verifiable and that the Mid-Fusion Ensemble model can be deployed reliably, we have streamlined the execution process. Consistency is the backbone of anomaly detection; therefore, we have taken specific measures to ensure that the results seen in this report can be replicated exactly.

6.1 Downloading the Dataset

To reproduce the final multimodal experiment, the target notebook is mid_fusion_mentalRoberta_wavLM.ipynb.

This notebook does not operate directly on the raw DAIC-WOZ ZIP files. Instead, it assumes that the dataset has already been downloaded, preprocessed, and that intermediate audio and text artifacts have been generated. The recommended workflow is as follows.

Dataset Preparation

Download the DAIC-WOZ dataset into the dataset/ directory. Ensure that the official split files are present:

- train_split_Depression_AVEC2017.csv
- dev_split_Depression_AVEC2017.csv
- full_test_split.csv

These files define the train, validation, and test partitions used throughout the project. All subsequent steps rely on these predefined splits.

Note, access to this dataset is at the discretion of University of South Carolina, and you would need to request access from them via email.

6.2 Environment Consistency and Seeding

One of the primary challenges in deep learning, especially when dealing with smaller, imbalanced datasets like our "Depressed" vs. "Control" classes, is stochasticity. To combat this, we have implemented a fixed seed across all libraries, including NumPy, PyTorch, and Python's native random module. By locking the seed, we ensure that the weight initialization, data shuffling, and dropout layers behave identically across different runs, guaranteeing that the 81% accuracy and 0.82 F1-score are not artifacts of a "lucky" initialization.

6.3 Running the Project

1. **File Structure and Path Verification**
   Before execution, the first and most critical step is to ensure that all required files, specifically the raw audio clips, text transcripts, and the configuration files, are placed in the correct directory paths as defined in the project structure. The notebooks are designed to look for datasets in a localized ./data folder; if these paths are not correctly mapped, the feature extraction process for WavLM and Mental-RoBERTa will fail to initialize.

2. **Execution via Jupyter Notebooks**
   We have consolidated the workflow into comprehensive notebooks that handle the pipeline from end to end. Running the notebooks will automatically trigger the following:

   - **Fixed Seed Initialization**: We use a fixed seed across all libraries (PyTorch, NumPy, and Python's random) to ensure that every run produces identical results. This eliminates variance in weight initialization and data shuffling, which is crucial for maintaining the integrity of our anomaly detection scores.

   - **Integrated Training and Evaluation**: Unlike workflows that separate these stages, our notebooks are scripted to perform the training epoch cycles followed immediately by a comprehensive evaluation on the test set. This ensures that the metrics reported, such as our 81% accuracy and 0.82 F1-score, are generated using the exact model state produced during that specific training session.

   By following this sequence, verifying paths, running the seeded notebooks, and utilizing the saved weights, we can faithfully reproduce the model's ability to detect anomalous depressed samples within the control group.

6.4 Step by step guide

Step 1: Isolate Participant-Only Audio

Run participant_audio_isolation.ipynb.

This step:

- reads each participant ZIP file in dataset/
- extracts *_AUDIO.wav and *_TRANSCRIPT.csv

- filters segments corresponding to the **Participant** using transcript timestamps
- stitches these segments into a single waveform

Output:

processed/isolated_audio/*.wav

Example:

300_P.zip → processed/isolated_audio/300_P.wav

Step 2: Generate Raw Transcripts

Run whisper_transcribe.ipynb.

This step:

- reads participant audio files from processed/isolated_audio/ or dataset/
- transcribes speech using Whisper model
- generates raw text transcripts

Output:

dataset/transcripts/

Step 3: Clean Transcripts

Run transcript_cleaning_minimal.ipynb.

This step:

- cleans Whisper-generated transcripts
- normalizes formatting
- removes noise for downstream text modeling

Output:

dataset/cleaned/

Step 4: Generate Audio and Text Feature Representations

Run the following notebooks:

- experiments/wavLM_MLP_cls.ipynb
- experiments/mental_roberta_lstm_cls.ipynb

These notebooks generate the unimodal feature caches used by the fusion model, and also train their respective classifiers, saving the best-performing checkpoints.

Audio Modality: WavLM Feature Extraction

Run wavLM_MLP_cls.ipynb.

This step:

- segments participant audio into 8-second windows with a 4-second hop
- extracts frozen WavLM embeddings for each segment
- stores segment-level audio features
- trains an audio classifier and saves the best checkpoint

Outputs:

- processed/wavlm_features_cls.npz
- experiments/best_model/wavlm_mlp_lstm_cls/wavLM_MLP_LSTM_cls_best.pth

Note:

The notebook first searches for *_P.wav files in dataset/. If not found, it falls back to processed/isolated_audio/, which is populated in Step 1.

Text Modality: Mental-RoBERTa Feature Extraction

Run mental_roberta_lstm_cls.ipynb.

This step:

- reads cleaned transcripts from dataset/cleaned/
- splits each transcript into overlapping 200-token windows
- extracts frozen Mental-RoBERTa [CLS] embeddings
- stores text feature representations
- optionally applies EDA-based augmentation
- trains a text classifier and saves the best checkpoint

Outputs:

- processed/mental_roberta_features_cls.npz
- processed/mental_roberta_features_eda_cls.npz *(optional)*
- experiments/best_model/mental_roberta_lstm_cls/mental_roberta_lstm_cls_best.pth

Step 5: Run the Final Multimodal Fusion Experiment

Run experiments/mid_fusion_mentalRoberta_wavLM.ipynb from start to finish.

This notebook performs the final multimodal training by combining the precomputed text and audio representations.

This step:

- loads cached Mental-RoBERTa text features
- loads cached WavLM audio features
- loads the best unimodal checkpoints
- freezes the unimodal encoders
- performs mid-level fusion of text and audio representations
- trains the final multimodal classifier
- evaluates performance on validation and test splits
- saves the best-performing fusion model

Required Inputs

Ensure the following artifacts from previous steps are available:

- processed/mental_roberta_features_cls.npz
- processed/mental_roberta_features_eda_cls.npz *(if EDA is used)*
- processed/wavlm_features_cls.npz
- experiments/best_model/mental_roberta_lstm_cls/mental_roberta_lstm_cls_best.pth
- experiments/best_model/wavlm_mlp_lstm_cls/wavLM_MLP_LSTM_cls_best.pth

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Outputs

- Fusion model checkpoints and evaluation outputs stored under:

  experiments/best_model/mid_fusion_mentalroberta_wavlm/

Summary

Once preprocessing, feature extraction, and unimodal training are complete, this step produces the final multimodal model by running the fusion notebook end-to-end.

6.5 Hardware Note

Given the computational requirements of extracting features from WavLM and Mental-RoBERTa simultaneously, we recommend running these notebooks on a system with a CUDA-enabled GPU. While the code will default to CPU if necessary, the inference time for the ensemble model is significantly optimized under GPU acceleration.

7. Challenges faced and Solutions

Throughout the course of this project, several key challenges were encountered, primarily relating to dataset limitations, computational constraints, and model generalization.

Dataset Size and Feature Complexity

The dataset size posed a significant limitation, particularly when working with high-dimensional feature spaces arising from segmented inputs (e.g., overlapping hops in audio processing). This resulted in a large feature space relative to the number of training samples, increasing the risk of overfitting and limiting the model's ability to generalize effectively.

Hardware and Computational Constraints

Training deep learning models, particularly transformer-based architectures, required substantial computational resources. On local hardware, projected training times were impractical, often exceeding two days for full experimentation cycles. To address this, we adopted a cloud-based workflow using Google Colab with A100 GPU acceleration, alongside Google Drive for data management. This significantly reduced iteration time and enabled more extensive experimentation.

Limitations of Pretrained Feature Extractors

Despite leveraging pretrained models trained on large-scale datasets, the extracted feature embeddings were not sufficiently discriminative for our task. In particular, standalone pretrained representations struggled to distinguish effectively between the two classes in our dataset.

To address this, we explored partial fine-tuning, unfreezing the upper layers (typically 2–4 layers) of the pretrained models. This allowed task-specific adaptation while retaining general representations. However, due to the limited dataset size and class imbalance, the benefits of fine-tuning remained constrained.

Final Approach: Transfer Learning and Integration

Given these limitations, our final approach relied more heavily on transfer learning strategies combined with model integration. Rather than depending solely on a single pretrained model, we incorporated multiple modalities and leveraged ensemble techniques to extract complementary information. This allowed the model to overcome the limitations of individual feature extractors and achieve improved performance.

8. Future Improvements

While our Mid-Fusion Ensemble has proven to be a reliable baseline for detecting anomalies, there are several architectural paths we can explore to further sharpen the model's sensitivity and precision.

8.1 Refined Attention Pooling Mechanism

Currently, our model uses Attention Pooling to aggregate features, but there is significant room to evolve this into a more discriminative learning process. In future iterations, we aim to implement more sophisticated attention layers that can better distinguish between "filler" segments and high-value emotional cues. By training the model to assign even higher weights to specific vocal tremors or linguistic markers, effectively "zeroing in" on the most discriminative features, we can improve the model's ability to catch the subtle anomalies that characterize the depressed class.

8.2 Multi-Model Embedding Stacks

Another promising direction is to move beyond a single feature extractor per modality. Instead of relying solely on Mental-RoBERTa or WavLM, we could employ multiple pretrained models for each modality (e.g., stacking embeddings from RoBERTa, word2vec, Mental-RoBERTa for text, alongside WavLM, wav2vec 2.0 and HuBERT for audio). By stacking these diverse embeddings, we create a richer feature space. This would allow us to perform intra-modality cross-feature learning, where the model learns to synthesize the unique strengths of different architectures before the data even reaches the fusion stage.

8.3 Advanced Cross-Modality Interaction

Finally, we see a massive opportunity in deeper cross-modality feature interaction. While our current mid-fusion approach concatenates features effectively, we plan to explore Cross-Attention mechanisms where the text modality can "query" the audio modality and vice-versa. This would allow the model to understand, for instance, how a specific choice of words in the text correlates with a specific change in pitch in the audio. Such a granular level of interaction could significantly boost the performance of the model in complex cases where the anomaly is only visible through the intersection of both signals.

By pursuing these possible enhancements, we aim to push the F1-score even higher and further reduce the rate of false positives, turning this classifier into an even more precise tool for behavioral health screening and anomaly detection.

9. Conclusion

This project began with an ambitious objective: to detect depressive anomalies directly from acoustic signals. While this formulation was theoretically appealing, our experimental results revealed that a purely audio-based approach was insufficient under the constraints of the available dataset. In this sense, the original problem was not fully solved in its initial form.

However, this limitation led to a critical insight. Depression is not expressed solely through how something is said, but also through what is said. By extracting textual information from the same audio source and combining it with acoustic features, we were able to construct a multi-modal representation that significantly improved detection performance. This shift from a rigid unimodal formulation to a more integrated approach proved to be the key turning point of the project.

A deeper analysis of the results suggests an asymmetry between the modalities. The textual modality, powered by Mental-RoBERTa, achieved strong performance even under limited data conditions, indicating that it captures a relatively stable and information-dense signal. In contrast, the audio modality exhibited weaker standalone performance, not due to a lack of relevance, but likely due to its higher data requirements. Acoustic patterns such as prosody, latency, and vocal variability are inherently more complex and require larger datasets to be learned effectively.

The modest yet consistent gains observed in the mid-fusion ensemble reflect this dynamic. Text provides a strong baseline signal, while audio contributes complementary information that becomes increasingly valuable when integrated. This suggests that while textual features may approach saturation at smaller scales, the contribution of audio features is expected to grow more significantly with increased data availability.

Ultimately, while the initial goal of purely audio-based anomaly detection was not achieved, the project successfully reframed the problem into a more realistic and effective paradigm. This evolution, from an overambitious starting point to a grounded, multi-modal solution, highlights the importance of aligning model design with both the structure of the data and the scaling behavior of different modalities. The resulting system demonstrates that meaningful progress in mental health anomaly detection lies not in isolating a single signal, but in leveraging the complementary strengths of both language and speech.

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